Disproof by Counter Example

Exam code: 9MA0

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Updated on 18 June 2025

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Disproof by counter example

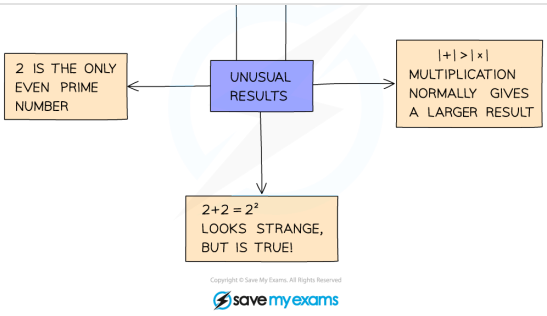
What is disproof by counter example?

- Disproof by counter examples involves finding a value that does **not** work for the given statement
  - That value is called a **counter example**

How do I disprove a result?

- You only need to find **one** value that does "not work"
- Numbers that have unusual results are often helpful to try

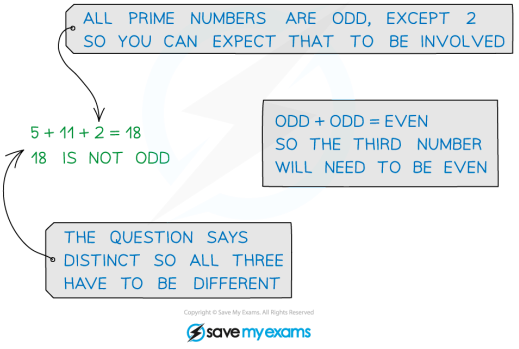
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



- Look carefully to see what types of number the result is claimed for – it may just be integers rather than real numbers
- Irrational numbers can be unusual
- Think about areas of maths where unusual things happen – such as in trigonometry where there are several angles with the same sine value

Worked Example

Use a counter-example to show that the sum of three distinct prime numbers is not always odd.



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